Barrett's oscillator-shuttle-circuits

Are Tesla's OSCs nonlinear beyond Kirchhoff and Ohm? — register entry H36

kronspice simulation report · 2026-11-07 · component values declared in the code (the sources give none)

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Summary

T. W. Barrett (*Ann. Fond. Louis de Broglie* 16, 23, 1991; US 5,493,691, 1996) says that Tesla's circuits, which shuttle energy among several floating “grounds”, are nonlinear in a way Kirchhoff's and Ohm's laws cannot describe. He says they mix a pump and a signal into an idler that obeys the Manley–Rowe relations, phase-conjugate the signal, and carry the algebra of quaternions (SU(2)) and higher Cayley–Dickson numbers. This example builds his Figs. 2–5 from the drawings, with every component value declared, then adds the nonlinear parts a real build has: a Metglas core and a spark gap.

- **The circuits as drawn are linear.** In all seven OSCs of Fig. 2 (A–G) the transient load current equals the AC superposition of pump and signal to ≤ 1.9×10^{-4}, the largest mixing product is ≤ 1.1×10^{-14} (round-off), and the ledger closes to ≤ 1×10^{-7}.
- **A nonlinear store makes the idler.** With a varactor top-load, an idler at fp − fs = 370 kHz appears at 4.063×10^{-5} of the pump line; Manley–Rowe sums are 4.0×10^{-7} and 4.8×10^{-7}; a +60° signal phase moves the idler to −62.71° (phase conjugation).
- **Standard scaling, not the paper's.** The idler goes as $Vs^{1.010}\cdot Vp^{1.027}$ (standard $\chi^{(2)}$ mixing, Es*·Ep); the paper's Eq. 24 has |Es|²·Ep.
- **Fig. 3 is a working pulse former:** load lobes −6.71, +3.42, −1.37 kV; ledger residual 1.0×10^{-10} J on 150.00 mJ stored.
- **The coil networks of Fig. 4 are ordinary two-port algebra:** cascades do not commute (XY vs YX 0.406) but are associative (1.1×10^{-10}) with det ABCD = 1, so they cannot carry Cayley non-associativity or zero divisors.
- **Real nonlinear parts do mix.** Air-core pancakes give no idler (≤ 6.91×10^{-15} V); a biased Metglas core gives a 1.09 mV idler; a timed spark gap multiplies a 1 V top-load signal into sidebands every 1 kHz (0.43× the signal rms, ordinary secondary; 0.66×, bifilar).

The claim and the standard prediction

Hypothesis (H36). OSCs, with coils shuttling energy among floating grounds, are nonlinear beyond Kirchhoff and Ohm; they mix pump and signal into a Manley–Rowe idler, phase-conjugate, and carry SU(2) and higher Cayley–Dickson algebra.

Standard prediction. Coils and capacitors, however their grounds float, form a linear network, and a linear network makes no new frequency. Parametric mixing, the Manley–Rowe relations and phase conjugation come from a nonlinear element (a varactor, a saturable core, a switch). Cascaded two-ports multiply as 2×2 matrices, which do not commute but do associate.

Neither the paper nor the patent gives a component value, so every value below is declared in the code. Spark gaps (Fig. 3) use the gap model and solver of *examples/sparkgap_diodebridge*; everything else runs in kronspice.

Fig. 2 — the OSCs A–G as drawn

Pump 10 V at 1 MHz from 10 Ω; pump coil α 10 µH, shuttle coil β 30 µH, load 50 Ω with its coil γ 5 µH, 2 Ω per coil; A's store 1 nF; each floating terminal a 10 pF top-load with a 1 GΩ leak; coupling k = 0.3 (0.6 for G's pancakes); F's capacitor 1 nF. The signal (1 V, 630 kHz) enters at the shuttle's terminal.

Fig.	load |I(fp)|	load |I(fs)|	transient vs AC (fp, fs)	largest mixing product	ledger
A	5.870×10^{-2} A	2.384×10^{-6} A	9.8e-05, 1.2e-05	9.7e-15 at (+3,+0)	7e-08

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Fig.	load |I(fp)|	load |I(fs)|	transient vs AC (fp, fs)	largest mixing product	ledger
B	6.209×10^{-2} A	1.041×10^{-3} A	5.5e-05, 1.7e-05	5.9e-15 at (−1,+3)	1e-08
C	5.865×10^{-2} A	4.951×10^{-6} A	1.3e-04, 2.2e-05	6.9e-15 at (+3,+3)	1e-08
D	5.832×10^{-2} A	1.450×10^{-5} A	7.9e-05, 4.3e-05	1.1e-14 at (+2,+3)	1e-08
E	5.461×10^{-2} A	1.457×10^{-6} A	1.4e-04, 2.1e-05	3.0e-15 at (+3,−1)	7e-09
F	3.508×10^{-2} A	1.541×10^{-6} A	1.9e-04, 4.3e-05	1.0e-14 at (+2,+0)	1e-07
G	5.853×10^{-2} A	9.894×10^{-6} A	6.8e-05, 4.8e-05	9.8e-15 at (+3,+1)	1e-08

Every circuit gives the pump and signal lines that AC analysis gives for each source alone, and no mixing product above round-off. **Bench test:** drive the pump, feed a signal to G2, and look for fp − fs in the load; the standard prediction is only what the parts' own nonlinearity gives (ferrite cores, ceramic dielectrics; air-core coils and film or vacuum capacitors give least).

With a nonlinear store (varactor)

Fig. 2A with the terminal top-load a varactor, q = Ct·(v + v²/(2·20 V)), pump 2 V, signal 0.2 V:

- idler at fp − fs = 370 kHz at the varactor terminal, 4.063×10^{-5} of the pump line;
- Manley–Rowe sums for the power into the varactor (relative, harmonics to 4th): 4.0×10^{-7} and 4.8×10^{-7} (zero for a lossless nonlinear reactance);
- idler proportional to $Vs^{1.010} \cdot Vp^{1.027}$: standard $\chi^{(2)}$ mixing, Es*·Ep. The paper's Eq. 24 (patent Eq. 15) has |Es|²·Ep, which would give exponent 2 in the signal;
- signal phase +60° moves the idler phase to −62.71° (the exact conjugate is −60°): phase conjugation.

The mixing, Manley–Rowe and phase conjugation Barrett describes are those of any parametric mixer; they come from the nonlinear element, not from the floating grounds.

Fig. 3 — switched −, +, − capacitors (spark gaps)

10 kV on 1 nF each, 1 µH into a 50 Ω load, 10 mm air gaps (breakdown 35.5 kV, so they hold the 20 kV between neighbours), triggered at 0, 100 and 200 ns.

quantity	value
load-voltage lobes, in order	−6.71, +3.42, −1.37 kV
strikes per gap	49, 3, 1
stored	150.00 mJ
to the load	66.27 mJ
in the gaps (arcs)	76.49 mJ
left in the capacitors	3.802 mJ
solver damping	3.442 mJ
ledger residual	1.0×10^{-10} J

This is a lumped frozen-wave generator, the impulse source of later ultrawideband radar. When a gap fires, the next capacitor shares charge with the one before it with no inductance between them; that loss goes to the arcs, which is why the + and second − lobes are smaller. Transmission-line segments in place of capacitors avoid it.

Fig. 4 — the coil networks (A quaternion, B Cayley, C sedenion-like)

α…ψ = 10…70 µH (1 mΩ each), 1 pF per node, 10 pF per open terminal, 5 MHz (declared). One-terminal networks are driven at their terminal against G1:

one-terminal network	Y / ($j\omega C_{term}$)	largest coil current / terminal current
A	$1.209029 - 1.60\times10^{-8}i$	1.73×10^{-1}
B(i)	$1.360694 - 5.44\times10^{-8}i$	2.65×10^{-1}
C(i)	$1.619236 - 1.80\times10^{-7}i$	3.82×10^{-1}

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With no capacitance at the inner nodes every coil current is zero: a network with one terminal is only its terminal's capacitance, so the coils (and the α, β, γ "products") act only through stray capacitance, or as antennas in an outside field.

two-ports X = B(ii), Y = C(ii), Z = B(ii) with values reversed	result
cascade XY in kronspice vs ABCD(X)·ABCD(Y)	1.9×10^{-16}
XY vs YX (order matters: "αβ ≠ βα")	0.406
XYZ vs (XY)Z and X(YZ) (associative)	1.1×10^{-10}
|det ABCD − 1| (reciprocal: invertible, no zero divisors)	1.1×10^{-16}
S†S − I (lossless: S unitary, an element of U(2))	1.8×10^{-6}

Order dependence is real and is ordinary two-port theory. The SU(2) Barrett names is the group of a lossless two-port's scattering matrix, up to phase. Box H's "(αβ)γ ≠ γ(αβ)" is a commutator, not an associator.

Fig. 5 — the quaternion "idler"

x = x0 + x1·i + x2·j + x3·k with x0 = Es*·$e^{-i\omega s\ t}$, x1 = Es·$e^{-i\beta s\ t}$, x2 = √Ep·$e^{i\omega p\ t}$, x3 = √Ep·$e^{-i\beta p\ t}$. Each component's line at fp − fs is 4.1×10^{-17}, 4.1×10^{-17}, 7.2×10^{-17}, 7.2×10^{-17} (relative): a sum of pump and signal holds no idler; one needs a product, i.e. a nonlinear element. x1, x3 and Eq. 24 also write β·t, but β is a propagation constant (rad/m), so β·t is not a phase; the travelling-wave phase is β·z.

Bench version of Fig. 2G: bifilar pancakes and a Metglas core

Pump coil α and shuttle coil β are the flat spirals of *pulsed_bifilar* (214 turns of 0.5 mm copper, 127 m): L 6.84 mH and 10.9 Ω each; self-capacitance 6.5 pF ordinary (first resonance 757 kHz) against 3.03 nF bifilar (34.9 kHz). The bench therefore runs below 35 kHz: pump 10 kHz 30 V, signal 6.3 kHz 1 V (50 Ω each) into a 10 nF store, idler 3.7 kHz, load 50 Ω, coupling k 0.6 (declared). Metglas 2605SA1 (declared): magnetizing inductance ×4, saturating from 20 mA, core Q 50; ribbon k_{me} 0.3, Q 300, 0.54 m long for 3.7 kHz.

case	load at fp	fs [V]	idler fp−fs	2fp−fs [V]	Manley–Rowe	ledger
ordinary pancakes, air	2.989 V	1.18e-02	8.41e-16 V	5.78e-16	—	3e-09
bifilar pancakes, air	2.571 V	1.22e-02	6.91e-15 V	1.01e-14	—	5e-09
bifilar, Metglas core	0.461 V	1.57e-02	1.41e-07 V	2.49e-03	2e-03, −2e-05	5e-04
bifilar, Metglas, DC bias	**0.595 V**	**1.59e-02**	**1.09e-03 V**	**1.81e-03**	**1e-04, −1e-04**	**2e-04**
bifilar, Metglas, bias, ribbon 3.7 kHz	0.656 V	1.54e-02	2.88e-05 V	1.84e-03	1e-04, −2e-04	3e-04

Values as printed in run_output.txt. The DC bias is 13 mA (README).

- **Air core:** ordinary or bifilar, the pancakes are linear and make no idler; the bifilar winding changes the self-capacitance 470×, so the lines differ, but adds no new frequency.
- **Unbiased Metglas:** the core is symmetric and makes only odd orders; the strongest product is the four-wave line 2fp − fs.
- **Biased Metglas:** a DC bias makes the even orders too, including the three-wave idler fp − fs. Manley–Rowe holds for the core's flux curve.
- **Ribbon tuned to the idler** cuts it 38× at the load: a magnetostrictive resonance is an impedance peak of the core branch, a trap at that frequency. To help the idler, tune it elsewhere.

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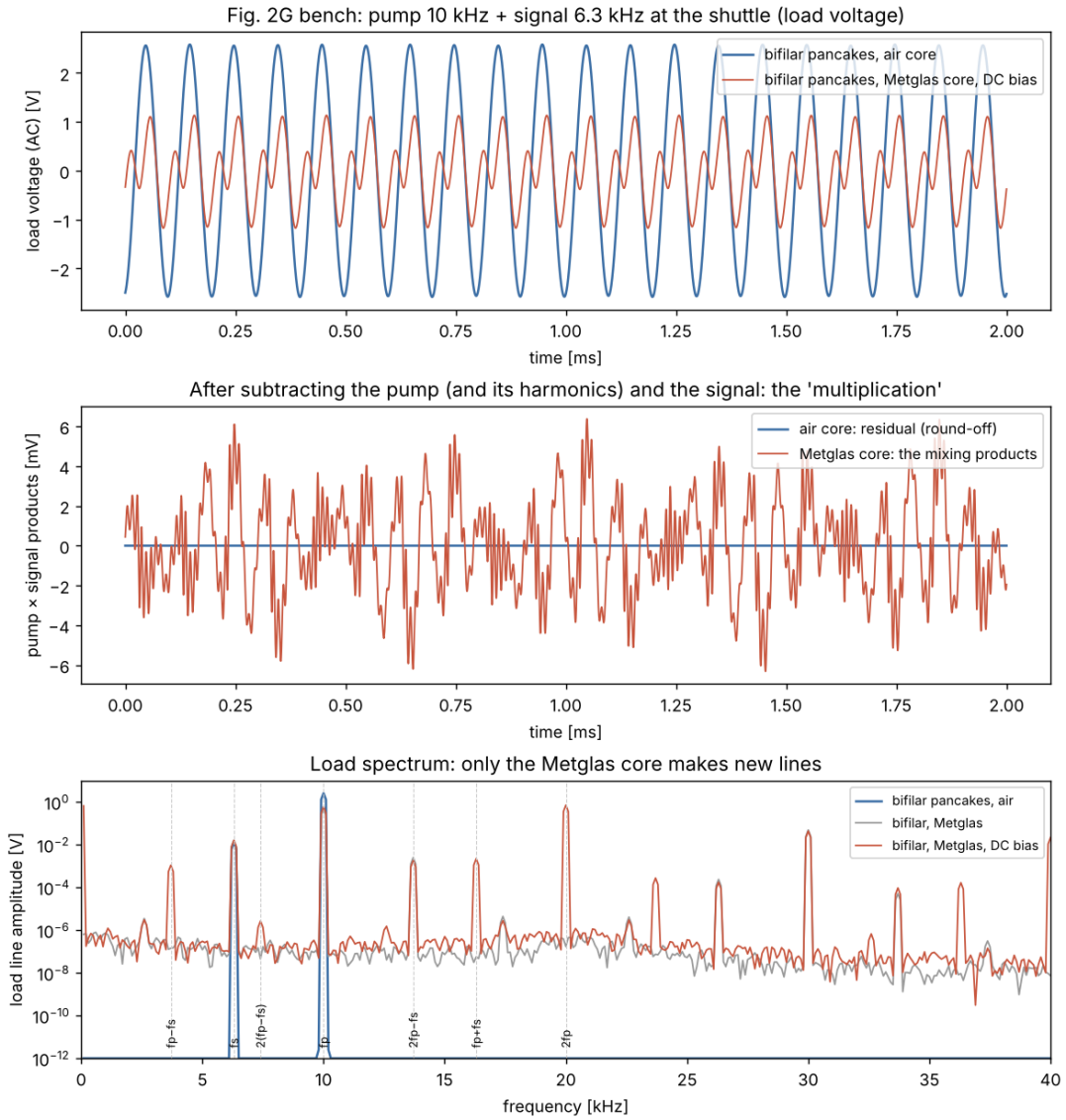


Figure 1. Fig. 2G bench. Top: load voltage with air-core bifilar pancakes and with a biased Metglas core. Middle: what is left after subtracting the pump (with harmonics) and the signal — round-off for air, a few mV of mixing products for the biased core. Bottom: load spectrum; only the Metglas core makes new lines (fp−fs, 2fp−fs, fp+fs, …). The ~10^{-6} V floor of the Metglas spectrum is numerical, from the core curve's piecewise-linear table.

A spark-gap Tesla coil with Barrett's floating terminal

Secondary: the 400-turn solenoid of *pulsed_bifilar*, L 4.85 mH, 15 pF top-load. Primary (declared): 10 kV DC through 10 kΩ charges 20 nF; a rotary gap fires 1000 times a second; k 0.15. A 1 V signal from 50 Ω enters in series with the top-load at 0.9 of the secondary's frequency. The gap is a timed switch (0.2 Ω for 100 µs per firing), so the coil is a linear, periodically switched circuit. The solver *tcsim.go* steps it by the trapezoidal rule; TestTeslaTwin checks it against kronspice with the gap held closed (top-load voltage within 0.1 %, ledger to 10^{-9}). Runs with the signal at phase 0 and π separate what is linear in the signal; subtracting the signal-alone run leaves what the switching does to it.

	ordinary solenoid	Tesla bifilar solenoid
self-capacitance; ring frequency	3.84 pF; 526.4 kHz	3.01×10^{3} pF; 41.5 kHz
primary Lp; signal frequency	4.6 µH; 473.5 kHz	735.1 µH; 37.5 kHz
top-load peak with sparks	289 kV	22 kV
signal alone at the top-load	3.38 V	0.022 V
sparks × signal, rms	1.02 V (0.43 × signal)	0.0103 V (0.66 × signal)
sidebands fs + k·1 kHz, k = −3…3 [V]	2.16e-01, 2.45e-01, 2.68e-01, 2.85e-01, 2.92e-01, 2.91e-01, 2.80e-01	3.22e-04, 4.13e-04, 5.29e-04, 6.86e-04, 9.20e-04, 1.34e-03, 2.50e-03
ledger residual	−4.7×10^{-10} J of 43.1 J	6.2×10^{-11} J of 49.5 J

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	ordinary solenoid	Tesla bifilar solenoid
gap heat; firings	7.77 J; 13 in 13 ms	0.276 J; 20 in 20 ms

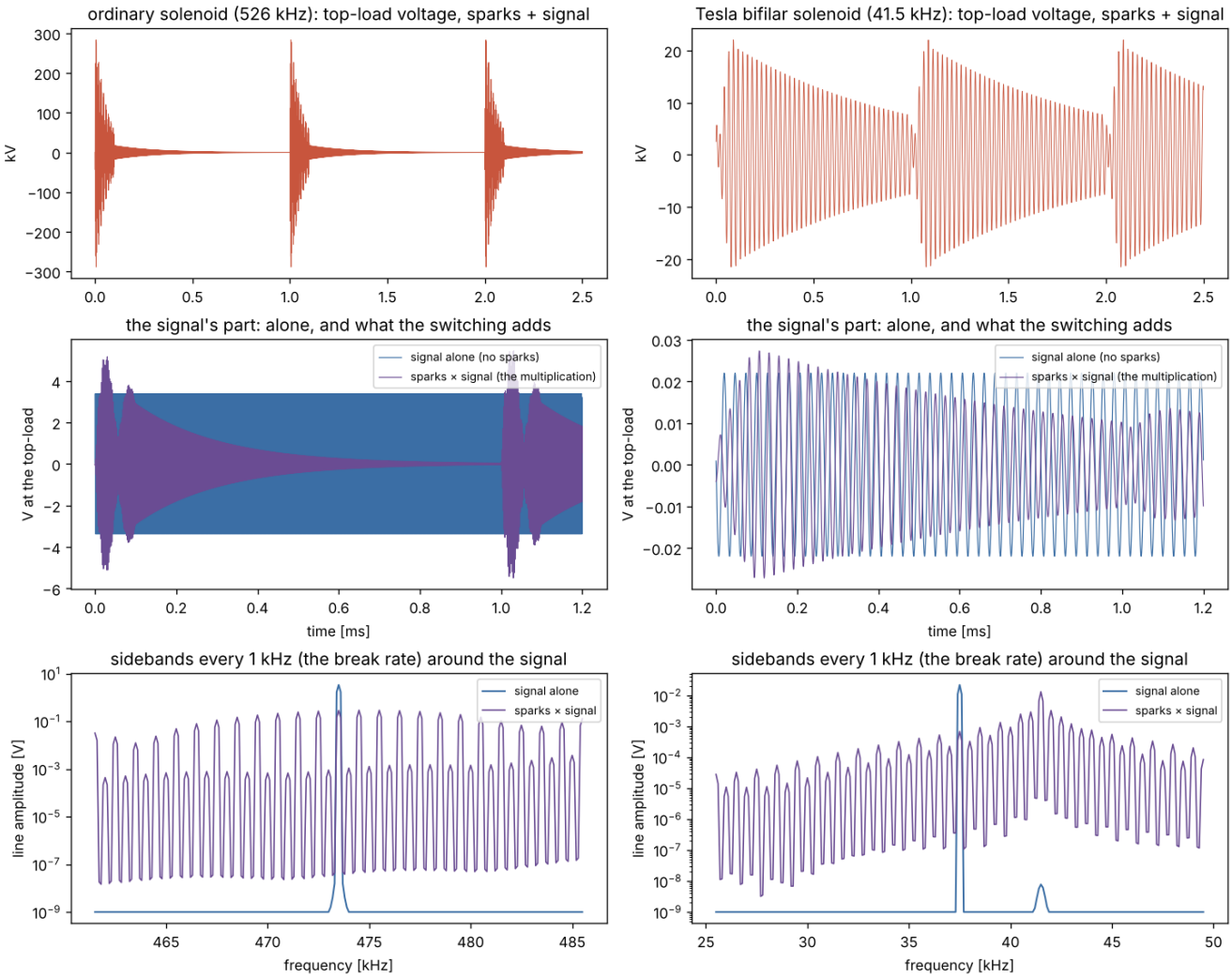


Figure 2. Spark-gap Tesla coil, ordinary secondary (left) and Tesla bifilar secondary (right). Top: top-load voltage with sparks and signal over three firings. Middle: the signal alone and the part the switching adds (sparks × signal). Bottom: spectra around the signal; the switching puts sidebands every 1 kHz (the break rate), flat for the ordinary coil and rising toward the bifilar coil's 41.5 kHz resonance.

This is real mixing, made by a time-varying element, of the kind Barrett's own Tesla coils would have shown. With the full arc model (strikes at thresholds, re-strikes while the channel is hot) the ordinary coil is hypersensitive: the phase-flip difference is 4.67×10^{3} V rms (1957× the signal at the top), with 1020 strikes in 13 ms. The 1 V signal flips individual re-strikes in a ring-down of hundreds of kV; a real gap's random jitter (ns–µs) swamps that, so it is not usable mixing.

Ledger and validation

- Fig. 2 ledgers 7×10^{-9} to 1×10^{-7}; transient equals AC superposition to ≤ 1.9×10^{-4}.
- Fig. 3 ledger residual 1.0×10^{-10} J of 150 mJ stored.
- Tesla-coil solver ledgers -4.7×10^{-10} J of 43.1 J and 6.2×10^{-11} J of 49.5 J; twinned with kronspice to 0.1 % (TestTeslaTwin).
- Metglas runs close to 2×10^{-4}–5×10^{-4}; air-core bench runs to 3×10^{-9}–5×10^{-9}.
- Spectra use a Hann window: a floating terminal charges slowly when a nonlinear store rectifies (leak time constant about a second), and the β–top-load ring has not quite died out after warm-up.

Limits of the model

- All component values are declared; the sources give none.

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- Coils are lumped equivalents (L, wire R, self-capacitance); the Metglas core is a piecewise-linear flux curve with a core-loss resistor, the ribbon a parallel tank in series with it.
- The rotary gap is a timed switch; the full arc model is used only to show the coil's sensitivity.
- Claims beyond these (noise rejection, lower transmission loss from “higher symmetry”) are untested here: the paper gives no measurement and no equation to simulate.

Status

Register status: **untested**. Figs. 2–5 as drawn are linear and obey Kirchhoff's laws; Barrett's mixing and phase conjugation are real effects of a nonlinear element, with standard scaling and the Manley–Rowe relations; the SU(2) and order effects are ordinary two-port algebra; Fig. 3 is a working pulse former. The bench test is fp − fs in the load of an air-core Fig. 2 build.

Reproducing the results

```
go run ./examples/barrett_osc                  # about 1.5 minutes
go run ./examples/barrett_osc -data DIR        # also the figure data (CSV)
python3 examples/barrett_osc/viz/plot_bench.py DIR viz/bench.png
python3 examples/barrett_osc/viz/plot_tesla.py DIR viz/tesla_coil.png
go test ./examples/barrett_osc
```


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